

# Nexora Engineering

Web Development • Electronics & Technology

## 1. Introduction

Nexora Engineering is a student-driven engineering portfolio and resource platform developed to present academic projects, engineering tools, technical reports, books, and learning resources in one organized website. The project combines a project showcase with an Engineering Library, providing a structured digital space for electronics, technology, and practical engineering work.

The website is built using HTML, CSS, and JavaScript. Its main sections include Home, Projects, Tools, Engineering Library, About, and Contact. The portfolio presents engineering projects together with the software and technologies used in their development.

## 2. Project Objectives

- Develop a professional student engineering portfolio.
- Showcase academic and practical engineering projects.
- Present the software and technologies used during project development.
- Provide an Engineering Library containing books, software resources, and useful websites.
- Organize the website through clear and simple navigation sections.
- Create a platform that can be expanded as additional engineering projects are completed.

## 3. Technologies Used

| Technology | Role in the Website                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------|
| HTML       | Creates the website structure, navigation, project cards, tool cards, library sections, and profile content. |
| CSS        | Controls layout, typography, spacing, cards, grids, buttons, and the overall visual presentation.            |
| JavaScript | Provides interactive behavior and supports the website interface.                                            |

Table 1 — Technologies used in the Nexora Engineering website.

## 4. Website Structure

The website is organized into separate sections so visitors can move from the main introduction to projects, engineering tools, learning resources, student information, and contact information. This section-based structure keeps the interface organized and makes the portfolio easy to navigate.

## Nexora Engineering Website Structure

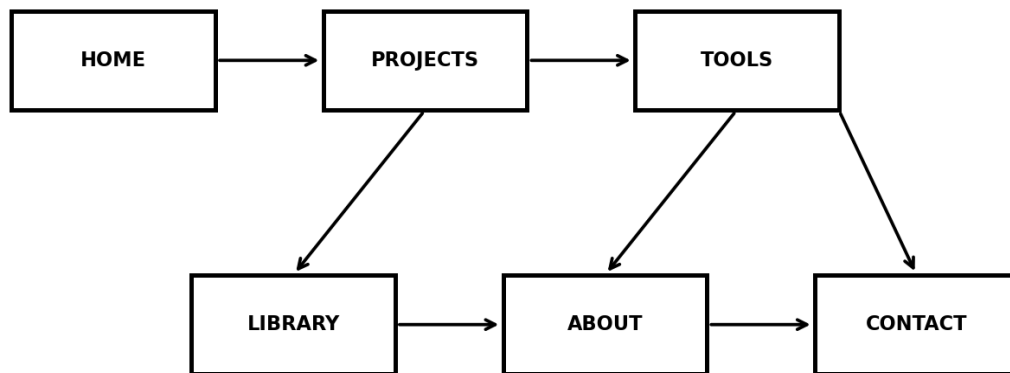


Figure 1 — Simplified structure of the Nexora Engineering website.

## 5. Home Section

The Home section introduces the platform using the title “Nexora Engineering” and the theme “Engineering Ideas Into Practical Solutions.” It presents the website as a student-driven platform focused on practical projects, technical tools, learning resources, and innovative solutions in electronics and technology.

The home area also provides a short introduction to the student portfolio and directs visitors toward the Projects and Engineering Library sections.

## 6. Projects Section

The Projects section is the main showcase of the portfolio. It presents engineering work from different areas including MATLAB and image processing, RF engineering, embedded systems, biomedical hardware, electromagnetics, electronics/PCB work, digital electronics, research, web development, and engineering resources.

| No. | Project                         | Category                       |
|-----|---------------------------------|--------------------------------|
| 01  | Brain Tumor Detection System    | MATLAB • Image Processing      |
| 02  | Wedge Antenna Design            | RF • Antenna Design            |
| 03  | Smart Water Quality Monitoring  | Embedded • Hardware            |
| 04  | Smart Patient Monitoring System | Biomedical • Hardware          |
| 05  | Eddy Current Braking System     | Electromagnetics • Hardware    |
| 06  | Transformer Design & Analysis   | Electronics • PCB              |
| 07  | Arithmetic Logic Unit           | Digital Electronics • Hardware |
| 08  | Eco Diplomacy                   | Research • Technology          |
| 09  | Engineering E-Library           | Web • Engineering              |
| 10  | Nexora Engineering              | Web Development                |

Table 2 — Projects presented in the Nexora Engineering portfolio.

### 6.1 Project Report Integration

Each project card can provide access to its full report through dedicated view and download actions. This makes the website both a visual project showcase and a document repository.

## 7. Engineering Tools and Technologies

The Tools section presents engineering software used across academic and practical work. These tools cover simulation, embedded systems, RF and antenna design, HDL verification, PCB development, and digital logic design.

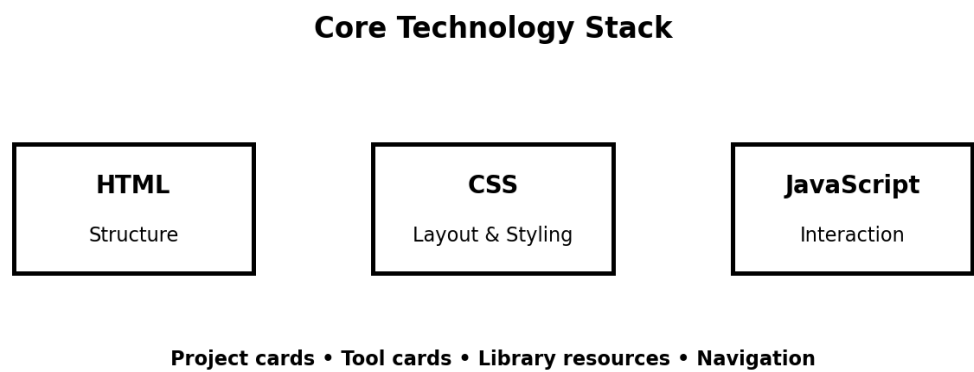


Figure 2 — Core web technologies supporting the Nexora Engineering interface.

| Tool         | Application                                                                              |
|--------------|------------------------------------------------------------------------------------------|
| MATLAB       | Signal processing, image processing, mathematical analysis, and engineering simulations. |
| STM32CubeIDE | STM32 programming, sensor interfacing, and embedded hardware projects.                   |
| LTspice      | Electronic circuit simulation and frequency-response studies.                            |
| CST Studio   | RF and antenna simulation/design.                                                        |
| ModelSim     | HDL simulation and digital logic verification.                                           |
| KiCad        | Schematic design and PCB layout.                                                         |
| Logisim      | Digital logic circuit design and testing.                                                |

Table 3 — Engineering tools included in the portfolio.

## 8. Engineering Library

The Engineering Library extends the portfolio into a learning-resource platform. It is organized around Engineering Books, Software & Downloads, and Useful Engineering Websites. The library is intended to keep technical learning material and project work within the same website.

## 9. About and Student Profile

The About section introduces the student portfolio and its engineering focus. It highlights interests in electronics, embedded systems, simulation, programming, and practical engineering projects.

## 10. Website Design and User Experience

The interface uses project cards, tool cards, categories, technology tags, navigation links, and report buttons to make information easy to scan. Short descriptions are shown first, while detailed reports and resources can be opened when required.

## 11. Advantages

- Combines an engineering portfolio with a learning-resource platform.
- Provides clear section-based navigation.
- Allows individual project reports to be accessed from project cards.
- Shows the engineering tools used across different projects.
- Provides a dedicated library for books, software, and technical resources.
- Can be expanded continuously with new projects and resources.

## **12. Current Limitations**

Some resource links may require final replacement with the actual report files, book files, software links, and external engineering websites before the website is deployed as a finished public platform.

## **13. Future Improvements**

- Connect every project card to its final verified report file.
- Add search and filtering for projects and library resources.
- Improve responsive behavior across mobile, tablet, and desktop screens.
- Add a database or content-management system for easier updates.
- Add a functional contact form.
- Improve accessibility and keyboard navigation.
- Allow filtering of projects by engineering discipline and software used.

## **14. Conclusion**

Nexora Engineering provides a structured digital platform for presenting student engineering projects alongside the tools and resources used to support them. The combination of project reports, engineering software, an Engineering Library, and student information gives the website the character of both a portfolio and a technical learning platform. Its HTML, CSS, and JavaScript foundation provides a practical base that can be expanded as new projects, resources, and interactive features are added.